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Electronic Circuit Design

Lecture notes

Chapter 2: Op-Amp Basics

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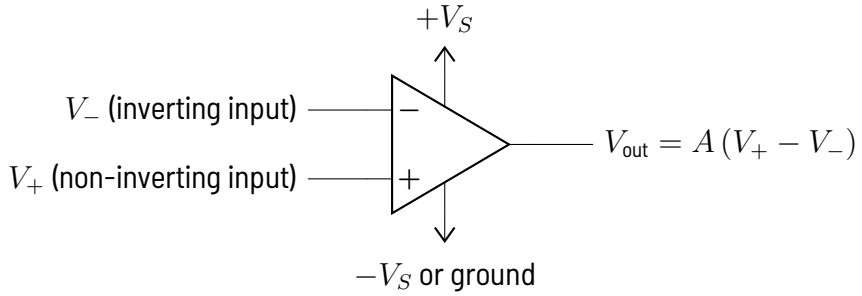
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2 Op-Amp Basics

2.1 The Op-Amp in a Nutshell

The operational amplifier (op-amp) appears in almost every chapter of this lecture: as a buffer behind a voltage reference, as the control element of a voltage or current source, as the amplifier behind a sensor bridge, in active filters and in the sample-and-hold circuit. This short chapter gives the minimum that the following chapters need: what an op-amp does, why it is always used with negative feedback, and the two golden rules that make the analysis of op-amp circuits easy. With these rules we look at the most useful op-amp circuit of all, the unity-gain buffer (voltage follower), and at the non-inverting amplifier.
Chapter 6 comes back to op-amps in detail: inverting, summing, difference and instrumentation amplifiers, and the limits of real op-amps. Good introductions are [1] (chapter 4) and [5].

Symbol and model



- an amplifier for the **difference** of two voltages, with a huge gain $A = 10^5 \ldots 10^7$
- inputs draw (almost) no current; the output can deliver some 10 mA
- the output stays inside the supply range; supply pins are often not drawn

The open-loop gain A is so large that a difference of a few hundred microvolts between the inputs is enough to drive the output to one of the supply rails. Without feedback the op-amp only tells which input is higher: it is a comparator (used in the flash ADC, chapter 9). To get a well-defined linear circuit, a part of the output voltage is fed back to the inverting input.

Negative feedback

- a part of V_{out} is fed back to the **inverting** input
- V_{out} rises $\Rightarrow$ V_- rises $\Rightarrow$ $V_+ - V_-$ shrinks $\Rightarrow$ V_{out} is pulled back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$ (50 µV for $V_{\text{out}} = 5$ V, $A = 10^5$)

Rule 2.1 Golden rules for an op-amp with negative feedback:

1. The output does whatever is necessary to make the voltage difference between the inputs zero: $V_+ = V_-$.
2. The inputs draw no current: $I_+ = I_- = 0$.

Rule 1 holds only while the feedback loop works: the output must not be stuck at a supply rail, and the feedback must go to the inverting input. Rule 2 is a property of the op-amp itself; real inputs draw picoamperes (CMOS) to some 10 nA (bipolar).

Real op-amps on a single supply

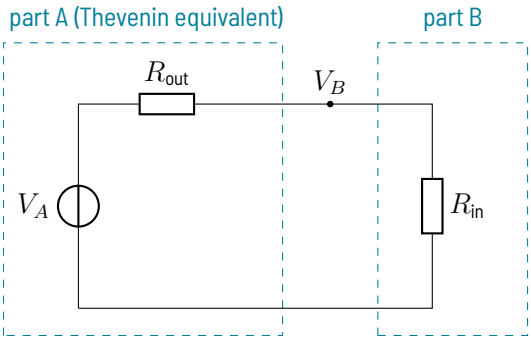
- most circuits in the lab run from a single supply (5 V or 12 V) and ground
- **input common-mode range**: the input voltages must stay inside it (often the supply minus 1.5 V)
- **output swing**: a rail-to-rail output reaches the rails within some 10 mV, older types (LM358) only within about 1.5 V of the positive rail
- **output current**: typically 10 mA … 40 mA, short-circuit protected
- examples: MCP6002, TLV9002 (CMOS, rail-to-rail, 5 V), LM358 (bipolar, up to 32 V)

The data sheet tells which voltages are allowed at the inputs and which output voltages are possible at a given load current. These two ranges are the first thing to check in every op-amp circuit, before any formula: the golden rules only hold inside them.

2.2 The Unity-Gain Buffer

A circuit is built from parts: a sensor divider, a filter, a reference, an amplifier, a converter input. Each part is designed for itself, with the formulas of chapter 1. When two parts are connected, the second part loads the first one, and the formulas are no longer valid: the voltage divider delivers less voltage, the filter has another cut-off frequency. The unity-gain buffer (voltage follower) separates the two parts. It copies the voltage from its input to its output, takes no current from the first part and delivers the current for the second part from the supply.

Every connection is a voltage divider



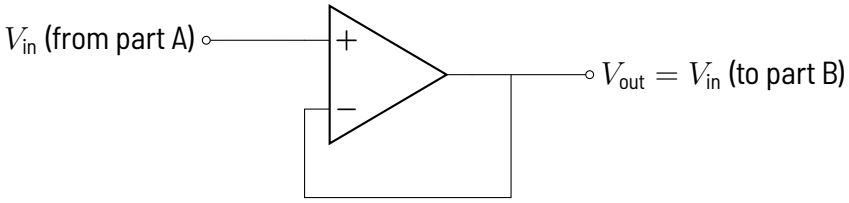
Loading

$V_B = V_A \dfrac{R_{\text{in}}}{R_{\text{in}} + R_{\text{out}}}$ error $\approx \dfrac{R_{\text{out}}}{R_{\text{in}}}$ for $R_{\text{in}} \gg R_{\text{out}}$

- rule of thumb: $R_{\text{in}} \geq 100\, R_{\text{out}}$ for an error below $1\,\%$

Part A is replaced by its Thevenin equivalent (chapter 1): a voltage divider of R_1 and R_2 has $R_{\text{out}} = R_1 \parallel R_2$, an RC low-pass has the output impedance $R \parallel (1/j\omega C)$. Part B is represented by its input resistance. If the condition $R_{\text{in}} \gg R_{\text{out}}$ cannot be met, a buffer is needed.

The voltage follower



- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$ (gain 1)
- rule 2: input current ≈ 0, input resistance GΩ …TΩ: part A is **not loaded**
- output resistance with feedback $r_o/(1+A)$: $50\,\Omega/10^5 = 0.5\,\text{m}\Omega$: part B sees an **ideal source**

Unity-gain buffer

$V_{\text{out}} = V_{\text{in}}$ $R_{\text{in}} \to \infty$ $R_{\text{out}} \to 0$

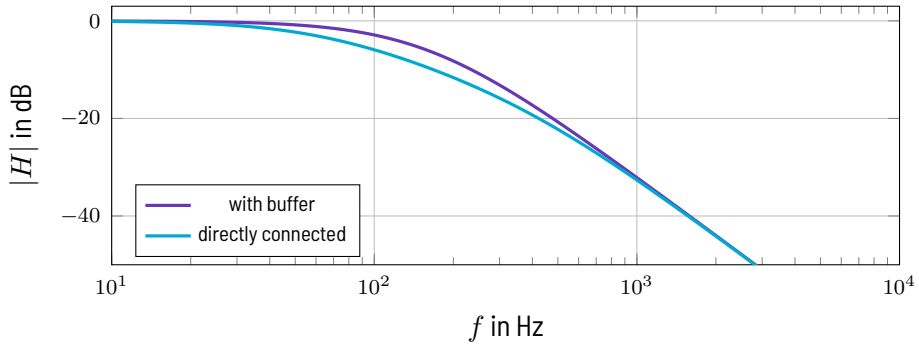
Example: potentiometer as a set-point

- potentiometer $10\,\text{k}\Omega$ at $5\,\text{V}$, wiper in the middle: $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$
- load $1\,\text{k}\Omega$ directly at the wiper: $V_B = 2.5\,\text{V} \cdot \frac{1}{1+2.5} = 0.714\,\text{V}$
- load behind a follower: $2.50\,\text{V}$; the op-amp delivers $2.5\,\text{mA}$
- the wiper itself still has $2.50\,\text{V}$: the follower does not load it

▶ Falstad: `examples/opampbasics/buffer_pot.txt`

The op-amp in the Falstad example runs from $5\,\text{V}$ and ground, like a CMOS rail-to-rail op-amp such as the MCP6002. With the slider of the directly connected load you can see how large R_L must be until the wiper voltage is close to $2.5\,\text{V}$. Exercise 2.1 works through this example.

Example: two RC low-passes in cascade



- each stage $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 159\,\text{Hz}$
- with buffer: $H = H_1 \cdot H_2 = \frac{1}{(1+j\omega RC)^2}$, at f_c: $|H| = 1/2$ (-6 dB)
- directly: $H = \frac{1}{1+3j\omega RC-(\omega RC)^2}$, at f_c: $|H| = 1/3$ (-9.5 dB)

▶ Falstad: `examples/opampbasics/buffer_rc_cascade.txt`

Without the buffer the second stage draws current from the capacitor of the first stage. The transfer function is no longer the product of the two single transfer functions, the cut-off frequency moves down and the corner becomes softer. The Falstad example drives both versions with a sine of 1 V amplitude at f_c: the scope shows 0.333 V for the direct and 0.500 V for the buffered cascade. With the buffer each stage can be designed on its own, which is the idea behind active filters (chapter 7).

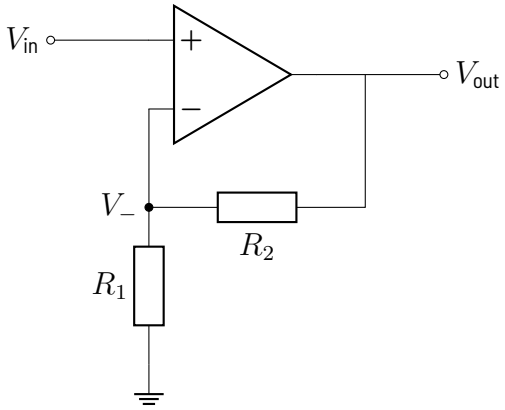
Where buffers are used – and their limits

- in this lecture: behind a voltage reference (chapter 4), at bridge taps (chapter 5), between filter stages (chapter 7), behind the hold capacitor (chapter 8)
- limits
  - output current (some 10 mA) and output swing
  - input common-mode range: e.g. LM358 inputs only up to $V_S - 1.5$ V
  - capacitive loads (long cables): the follower may oscillate; add $50\,\Omega$ ... $100\,\Omega$ in series with the output
  - offset voltage: $V_{\text{out}} = V_{\text{in}} + V_{\text{OS}}$ (µV ...mV)

2.3 The Non-Inverting Amplifier

If only a part of the output voltage is fed back, the follower becomes an amplifier. It keeps the good input property of the follower: the source is not loaded. The next chapter uses it to scale a reference voltage into an exact output voltage.

Non-inverting amplifier



- R_1, R_2 divide the output: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$
- $R_2 = 0$ (or $R_1 \to \infty$): the follower with $G = 1$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$, output resistance ≈ 0

The gain depends only on the ratio of two resistors, not on the op-amp. With $1\,\%$ resistors the gain is accurate to about $2\,\%$; with $0.1\,\%$ resistors to $0.2\,\%$. Chapter 6 derives the inverting amplifier and shows how a finite open-loop gain reduces the accuracy.

Exercise 2.1: Buffering a potentiometer

A $10\,\text{k}\Omega$ potentiometer between $5\,\text{V}$ and ground sets a voltage. The wiper is in the middle position. The voltage is used by a following stage with an input resistance of $R_L = 1\,\text{k}\Omega$.

a) Draw the Thevenin equivalent of the wiper. In which wiper position is R_{out} largest?

b) Calculate the voltage at R_L without a buffer and the error in percent.

c) How large must R_L be for an error below $1\,\%$ without a buffer?

d) Now a voltage follower (op-amp at $5\,\text{V}$, rail-to-rail) is placed between wiper and R_L. Calculate the voltage at R_L, the current from the wiper and the output current of the op-amp.

e) The wiper is turned to the upper end ($5\,\text{V}$). What does an LM358 at $5\,\text{V}$ supply deliver, and why?

f) Check (b) and (d) with Falstad.

Solution 2.1

a) $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$. For a wiper position x (0 ...1): $R_{\text{out}} = x(1-x)\,10\,\text{k}\Omega$, largest in the middle ($R/4$).

b) $V_B = 2.5\,\text{V} \cdot 1\,\text{k}\Omega/(1\,\text{k}\Omega + 2.5\,\text{k}\Omega) = 0.714\,\text{V}$, error $-71\,\%$.

c) $R_L \geq 100 \cdot 2.5\,\text{k}\Omega = 250\,\text{k}\Omega$.

d) $V_B = 2.50\,\text{V}$; wiper current ≈ 0 (picoamperes for a CMOS op-amp); op-amp output current $2.5\,\text{V}/1\,\text{k}\Omega = 2.5\,\text{mA}$, which comes from the $5\,\text{V}$ supply.

e) The input common-mode range of the LM358 ends at about $V_S - 1.5\,\text{V} = 3.5\,\text{V}$, and its output does not reach the positive rail either: the output stays at about $3.5\,\text{V}$. For signals up to the supply a rail-to-rail op-amp is needed.

f) ▶ Falstad: `examples/opampbasics/buffer_pot.txt`

`V_direct` $= 0.714\,\mathrm{V}$, `V_wiper` $=$ `V_buffered` $= 2.50\,\mathrm{V}$.

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